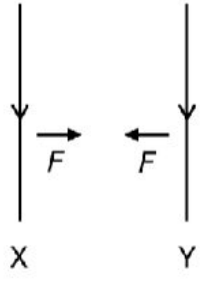
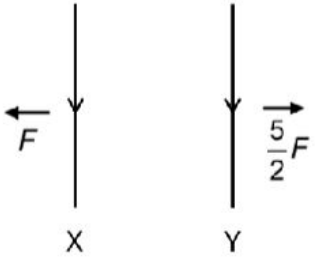
22 Two long, straight, current-carrying wires, X and Y, are parallel. The current in wire X is 2 A and the current in wire Y is 5 A.

Which diagram shows the correct combination of the magnitude of the forces and the directions of the current in wires X and Y?

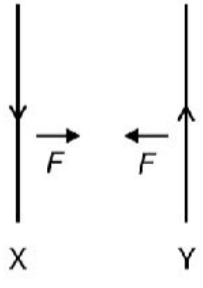
A



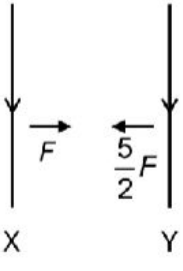
B



C



D



Force magnitude:

By Newton's Third Law, the forces on X and Y must be **equal in magnitude** — so both forces must be *F* (not 5/2 *F* on one and *F* on the other). This eliminates **B and D**.

Direction of forces:

- Currents in the **same direction** → wires **attract** (forces point toward each other)
- Currents in **opposite directions** → wires **repel** (forces point away from each other)

Checking A vs C:

Circuit **A**: Both currents flow downward (same direction) → wires should attract → forces point toward each other (X's force points right →, Y's force points left ←). This matches Circuit A. ✓

Circuit **C**: X flows downward, Y flows upward (opposite directions) → wires should repel → forces point away from each other. But Circuit C shows forces pointing toward each other — contradiction. ✗

Ans: A